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 Introduction to Module 6's Study
Video • 1 min

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Video • 1 min

Particle Swarm Optimization: A First Introduction
Video • 6 min

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✔ Particle Swarm Optimization: Operators
Video • 7 min

✔ Particle Swarm Optimization: Workflow Demo
Video • 11 min

Materials on Particle Swarm Optimization
Reading • 10 min

✓ **Understanding Particle Swarm Optimization Mechanics**
Reading • 5 min

- Use PSO via MATLAB: PSO Function
Video • 6 min
- Use PSO via MATLAB: Employ the Function
Video • 3 min
- Materials on Using Particle Swarm Optimization
Reading • 10 min

Use PSO via MATLAB: PSO Function
Video • 6 min

Use PSO via MATLAB: Employ the Function
Video • 3 min

Materials on Using Particle Swarm Optimization
Reading • 10 min

- Use PSO via MATLAB: Case Studies
Video • 6 min
- Materials on Using Particle Swarm Optimization
Reading • 10 min
- Applying PSO to a Real Engineering System
Reading • 5 min

Use PSO via MATLAB: Case Studies
Video • 6 min

Materials on Using Particle Swarm Optimization
Reading • 10 min

Applying PSO to a Real Engineering System
Reading • 5 min

Module 6 Quiz
Practice Assignment • 30 min

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Practice Assignment • 30 min

Module 6 Assignment
Graded App Item • 30 min

Objective function for Assignment 6
Reading • 5 min

Designing and Applying Particle Swarm Optimization to Solve an Engineering Optimization Problem
Dialogue • 5 min

Module 6 Recap
Reading • 5 min

Keeping Learning with UofG Online
Reading • 5 min

Module 6 Assignment
Graded App Item • 30 min

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- How does PSO differ from genetic algorithms in terms of how new solutions are generated?
- In your own words, what do **pBest** and **gBest** represent, and why are both necessary for effective optimization?
- How do the velocity and position update equations enable particles to move toward better solutions without explicitly using crossover or mutation?
- What roles do the parameters c_1 , c_2 , random factors r_1 , r_2 , and inertia weight ω play in balancing exploration and convergence?
- Based on the workflow demo, why does PSO tend to gradually cluster particles around promising regions of the search space?

Go to next item ✓ Completed

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